

ROHAN SANJAY SHINDE

Hardware Design Engineer | EMI/EMC Testing | Embedded Firmware | Power Supply Design

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PROFESSIONAL SUMMARY

Hardware Design Engineer with 4.5 years of experience specialising in industrial EMI/EMC & safety compliance testing (IEC 61326-1) and isolated power supply design (PI Expert). Skilled in board-level development, microcontroller port assignment, and component selection through rigorous datasheet analysis, spanning the full product lifecycle from circuit design through production deployment. Experienced in Root Cause Analysis (RCA), Corrective Actions (CA), and Preventive Actions (PA) for hardware failures on the production line, and in component cost-reduction drives and circuit standardisation. Collaborated with customer support and BD teams to resolve field issues. Currently building a Python and Streamlit-based document verification tool to automate the company's PDF audit process, targeting a reduction of over 40 hours/month of manual effort. Complements this RnD & production experience with self-directed embedded firmware projects on STM32 and ESP32 platforms, including bare-metal driver development and hardware-verified power electronics control.

TECHNICAL SKILLS

Hardware	Board-level debugging, EMI/EMC & Safety testing (IEC 61326-1), relay circuits, MOV protection, optocoupler isolation, power supply design (PI Expert)
Microcontrollers	Renesas RL78 series (R7F0C010B), RA series (R7FA0E1073CFJ) — port assignment, GPIO configuration, basic peripheral setup; STM32F103 and ESP32 (personal projects)
Embedded Firmware	Register-level bare-metal C (STM32, CMSIS), Arduino/C++ (ESP32), timer/PWM-based switching control, I2C, WiFi/HTTPS integration (personal projects)
IDEs & Tools	e ² Studio, CubeSuite+, VS Code, JIRA, PI Expert, arduino-cli, CMake, LTspice, KiCad (basic), CorelDraw (basic)
Communication	UART, RS-485 (automation purpose), I2C, WiFi/HTTPS
Programming	Ladder Logic (PLC), Python (prompt engineering & scripting — beginner), Embedded C (bare-metal drivers, port definitions)
Test Equipment	Digital Oscilloscope, Multimeter, Logic Analyzer
Process & Quality	RCA / CA / PA, CAPA, BOM analysis, datasheet evaluation, SMT & wave soldering process knowledge

WORK EXPERIENCE

Selec Controls Pvt. Ltd. | Design Engineer | Mumbai | 2022 – Present

April 2022 – July 2023	
Delivery	9 Products delivered: XT546A, XT543A, XT246A, XT346A, XT56A, XT264A, XT364A, XC10, XC410C — Timer/Counter series revamp.
Contributions	– Project approval documentation for all 9 products – Approximate costing and power consumption calculations for 4 products – CAD input including circuit diagram for 4 products – Board-level testing for 4 products – EMI/EMC & Safety testing for 2 products – Production deployment for 4 products – Microcontroller port assignment (Renesas RL78 – R7F0C010B); basic hardware definition file editing
Skills Gained	Datasheet analysis, EMI/EMC & Safety testing guidelines, circuit finalisation, mass production deployment, CubeSuite+ environment familiarisation

August 2023 – July 2024	
Delivery	Debug clearance, quality improvement & customer issue resolution across existing product range.
Contributions	– Cleared backlog of pending debug BINs for older products – Performed Root Cause Analysis (RCA); raised Corrective Actions (CA) and Preventive Actions (PA); responded to 5-Why analysis from quality team – Resolved JIG and PLC wiring issues to ensure smooth production – Handled critical customer complaints requiring new operational solutions – Observed and understood SMT, pick-and-place, and wave soldering machines on the production floor
Skills Gained	Hands-on debugging, production floor process understanding (SMT / pick-and-place / wave soldering), customer issue management, basic CorelDraw

August 2024 – August 2025	
Delivery	Component cost-reduction drive and hardware block standardisation.
Contributions	– Validated existing and alternate components against datasheet specifications (mechanical fitment, temperature ratings, operating parameters) – Contributed to hardware block standardisation by analysing circuits using DSO and providing supporting calculations – Proposed a new product concept; proposal deferred due to business focus constraints – Coordinated with sales team for basic market understanding
Skills Gained	Thorough datasheet evaluation, BOM cost-reduction methodology, Ladder Logic (PLC), RC-based circuit analysis for timer/counter series

September 2025 – March 2026	
Delivery	CAD release for End-of-Life (EOL) products and isolated Power Supply design.
Contributions	– Released circuit diagrams for EOL products – Designed an isolated Power Supply from scratch using PI Expert; collaborated with an external power electronics expert from Power Integrations for transformer finalisation – Performed microcontroller port assignment for the new product circuit (RA series) – Used e² Studio for minor code modifications
Skills Gained	Isolated PS design (PI Expert), transformer requirement specification, e ² Studio usage, basics of Embedded C

April 2026 – Present	
Delivery	Automation of company-wide PDF audit process (ongoing).
Contributions	– Developing an automated document verification tool using Python and prompt engineering to replace manual PDF auditing, targeting a reduction of 40+ hours/month of manual effort – Building and presenting auditing utilities via Streamlit for internal operational use – Currently automating EMI/EMC summary report to reduce manual effort and errors
Skills Gained	Prompt engineering, Python scripting (beginner), Streamlit, process automation concepts

PERSONAL PROJECTS

Claude Meter — ESP32 Real-Time Usage Display		2026
Project	Personal embedded systems project github.com/ROHANSHINDE1234/Claude_meter	
Highlights	– Designed and built firmware for an ESP32-WROOM-32 (Arduino/C++) driving an SH1106 I2C OLED display to show live Claude Code API usage, using a CLI-only toolchain (arduino-cli) with no vendor IDE – Implemented direct WiFi + HTTPS integration to poll usage data from API response headers, removing the need for a relay device or companion laptop daemon – Engineered secure, reflash-free credential management: NVS flash storage for persistent tokens, a password-gated local web server for updating tokens in the field, and a QR-code boot screen for one-scan mobile access – Built a modular, interface-driven architecture (IScreen, IUsageProvider, DisplayManager) cleanly separating display rendering, usage-state management, and network/API logic – Independently diagnosed and resolved hardware/firmware issues, including a display driver mismatch (SH1106 vs. SSD1306), a retired API model identifier, and a retry-pacing bug in the polling loop – Validated the complete system end-to-end on hardware: WiFi connection, live HTTPS calls, NTP time sync, and correct real-time data rendering on the OLED	
Tech	ESP32, arduino-cli, C++, I2C, WiFi/HTTPS, NVS flash storage, QR code generation	

Invertor — Solar-Powered Backup Inverter System		Ongoing
Project	Personal power electronics project github.com/ROHANSHINDE1234/Invertor	
Highlights	– Designing and building a solar-powered backup power system from first principles: solar array → MPPT charge controller → LiFePO4 battery/BMS → STM32-driven inverter → 230V AC output – Developed bare-metal, register-level STM32F103 ("Blue Pill") firmware for a two-stage inverter: a 25kHz center-aligned push-pull PWM driver (TIM4) and a 50Hz H-bridge output unfold (TIM3) – Redesigned the push-pull gate-drive scheme from a software/ISR-based approach to dual hardware-channel PWM, eliminating a timing race condition that risked MOSFET shoot-through – Implemented a 4-phase H-bridge state machine with enforced dead-time at every polarity transition to prevent cross-conduction – Verified all gate-drive timing and dead-time on a logic analyzer, confirming zero shoot-through across both power stages – Authored the transformer design specification and power-stage schematic (LTspice); planned PCB layout approach (KiCad, 2-layer) and full bill of materials	
Tech	STM32F103, C (bare-metal/register-level), CMSIS, CMake, ARM GCC toolchain, LTspice, KiCad, logic analyzer	

EDUCATION

Bachelor of Engineering – Electronics & Telecommunication | 71.78%

University of Mumbai | 2017 – 2021

CURRENTLY LEARNING

- Open to hardware design opportunities in industrial EMI/EMC-compliant applications.
- Exploring Embedded C further — working towards understanding RTOS concepts and communication protocols.